

Week 3 - Day 7

Customer Churn Prediction - Final Project

AIM2 Preparation | Week 3 Machine Learning | 7/7 COMPLETE

Project Goal

Build an end-to-end classification workflow for predicting telecom customer churn, combining Week 2 data analysis skills with Week 3 machine learning skills.

Target: 0 = customer stays, 1 = customer churns.

1. Project File Structure

```
AIM2-Preparation/  
|-- week-03-machine-learning/  
    |-- notes/  
    |-- notebooks/  
    |-- datasets/  
    |-- customer-churn-prediction/  
        |-- customer_churn.csv  
        |-- customer_churn_prediction.py  
        |-- confusion_matrix.png  
        |-- Plot feature importance.png  
        |-- Day 7 project notes PDF
```

2. End-to-End Workflow

Business Problem -> Load -> Inspect -> Clean/Validate -> EDA -> Define X/y -> Train/Test Split -> Train Models -> Evaluate -> Compare -> Confusion Matrix -> Feature Importance -> Business Recommendation -> Predict New Customers

3. Data Inspection & EDA

Item	Result
Dataset shape	60 rows x 7 columns
Missing values	0
Duplicate rows	0
Stay (0)	30 customers - 50%
Churn (1)	30 customers - 50%
Target balance	Perfectly balanced

Churn vs Stay Averages

Churn	Monthly Charges	Contract Months	Support Calls	Tenure Months	Data Usage GB
Stay (0)	58.43	14.80	1.63	33.10	32.57
Churn (1)	96.23	3.33	4.23	11.40	68.27

Observed associations in this synthetic dataset: churners had higher average monthly charges, shorter contracts, more support calls, shorter tenure, and higher data usage. These are associations, not causal conclusions.

Customer_ID was excluded because it is an identifier rather than a meaningful predictive customer attribute.

4. Train/Test Split

Features: Monthly_Charges, Contract_Months, Support_Calls, Tenure_Months, Data_Usage_GB. Target: Churn.

Split: test_size=0.25, random_state=42, stratify=y. Training samples = 45; testing samples = 15. Stratification preserves the target-class proportions.

5. Models & Final Evaluation

Model	Train Acc.	Test Acc.	Precision	Recall	F1
Logistic Regression	1.000	0.933	0.875	1.000	0.933
Decision Tree (depth=3)	1.000	0.933	0.875	1.000	0.933
Random Forest (100 trees)	1.000	1.000	1.000	1.000	1.000

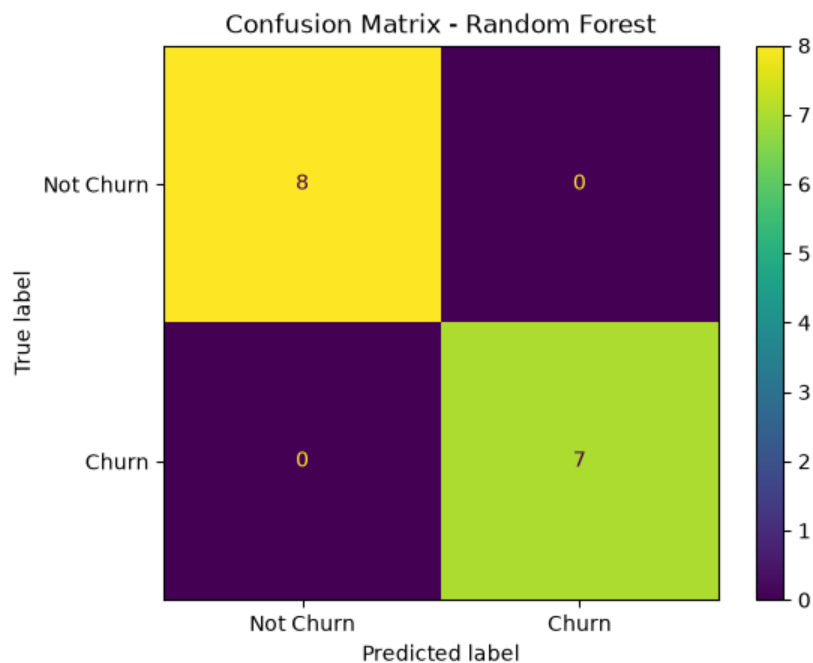
Highest Accuracy: Random Forest. **Highest Precision:** Random Forest. **Highest Recall:** all three models tie at 1.00. **Highest F1:** Random Forest.

Because the business priority is avoiding missed churners, Recall is especially important. All three models had perfect Recall on this test set, but Random Forest also had perfect Precision, Accuracy, and F1.

6. Confusion Matrices

Model	TN	FP	FN	TP
Logistic Regression	7	1	0	7
Decision Tree	7	1	0	7
Random Forest	8	0	0	7

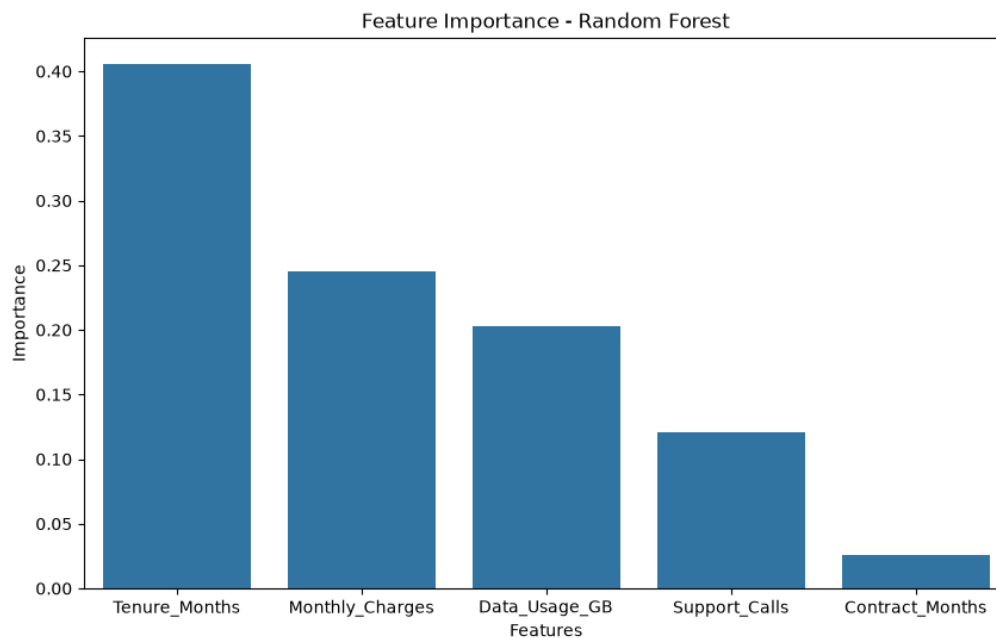
All models produced FN=0, so none missed an actual churner in this particular test set. Logistic Regression and Decision Tree each produced one false positive; Random Forest produced none.



7. Random Forest Feature Importance

Feature	Importance
Tenure_Months	0.405655
Monthly_Charges	0.245114
Data_Usage_GB	0.202635
Support_Calls	0.121008
Contract_Months	0.025588

Tenure_Months had the highest feature importance. This means the fitted Random Forest relied on it more than the other features when making predictions. Feature importance does not tell us the direction of the relationship and does not prove causation. Separate EDA showed that churners had shorter average tenure.



8. Business Recommendation

Recommended model for this experiment: **Random Forest**. All three models achieved Recall=1.00, satisfying the main goal of not missing actual churners. Random Forest additionally achieved Accuracy=1.00, Precision=1.00, and F1=1.00, with FP=0 and FN=0.

The dataset contains only 60 synthetic customers and a 15-customer test set. Perfect test performance should therefore be treated as a learning result, not evidence of guaranteed real-world performance. A production decision would require larger, representative real-world data and further validation.

9. New Customer Predictions

Customer	Prediction	Churn Probability
Customer A	Stay	0.0000
Customer B	Churn	1.0000
Customer C	Churn	1.0000

A model probability of 1.0000 on this synthetic exercise should not be interpreted as certainty that a real customer will churn.

10. Week 3 Final Assessment - Revision

Feature: input used for prediction. **Target:** output the model predicts.

Regression: predicts continuous values. **Classification:** predicts categories.

Train/test split: trains on one portion and evaluates generalization on unseen data.

Overfitting: model learns training-specific noise/details and performs worse on unseen data.

Precision: trustworthiness of positive predictions. **Recall:** proportion of actual positives found.

Imbalanced data: Accuracy can be misleading when a majority-class prediction dominates.

Decision Tree vs Random Forest: one tree versus an ensemble of randomized trees.

Feature importance: model reliance, not causation.

Model selection: evaluate unseen performance and business-relevant metrics, not training Accuracy alone.

11. Week 3 Completion Summary

Day	Topic	Status
1	Machine Learning Fundamentals	Complete
2	Linear Regression & Model Evaluation	Complete
3	Classification & Logistic Regression	Complete
4	Classification Model Evaluation	Complete
5	Decision Trees & Overfitting	Complete
6	Random Forest & Model Comparison	Complete
7	Customer Churn Prediction Final Project	Complete

WEEK 3 - MACHINE LEARNING: 7/7 COMPLETE

Final workflow mastered: Data inspection -> EDA -> feature/target definition -> train/test split -> model training -> evaluation -> confusion matrices -> model comparison -> feature importance -> business reasoning -> predictions.